Ksp FRQ #1

1. Answer the following questions relating to the solubility of the chlorides of silver and lead.

   (a) At 10°C, 8.9×10^{-5} g of $AgCl(s)$ will dissolve in 100. mL of water.

   (i) Write the equation for the dissociation of $AgCl(s)$ in water.

   (ii) Calculate the solubility, in mol L^{-1}, of $AgCl(s)$ in water at 10°C.

   (iii) Calculate the value of the solubility-product constant, K_{sp}, for $AgCl(s)$ at 10°C.

   (b) At 25°C, the value of K_{sp} for $PbCl_2(s)$ is 1.6×10^{-5} and the value of K_{sp} for $AgCl(s)$ is 1.8×10^{-10}.

   (i) If 60.0 mL of 0.0400 *M* $NaCl(aq)$ is added to 60.0 mL of 0.0300 *M* $Pb(NO_3)_2(aq)$, will a precipitate form? Assume that volumes are additive. Show calculations to support your answer.

   (ii) Calculate the equilibrium value of $[Pb^{2+}(aq)]$ in 1.00 L of saturated $PbCl_2$ solution to which 0.250 mole of $NaCl(s)$ has been added. Assume that no volume change occurs.

Ksp FRQ #2

1. Answer the following questions relating to the solubilities of two silver compounds, Ag_2CrO_4 and Ag_3PO_4.

   Silver chromate dissociates in water according to the equation shown below.

   $$Ag_2CrO_4(s) \rightleftarrows 2\,Ag^+(aq) + CrO_4^{2-}(aq) \qquad K_{sp} = 2.6 \times 10^{-12} \text{ at } 25°\text{C}$$

   (a) Write the equilibrium-constant expression for the dissolving of $Ag_2CrO_4(s)$.

   (b) Calculate the concentration, in mol L^{-1}, of $Ag^+(aq)$ in a saturated solution of Ag_2CrO_4 at 25°C.

   (c) Calculate the maximum mass, in grams, of Ag_2CrO_4 that can dissolve in 100. mL of water at 25°C.

   (d) A 0.100 mol sample of solid $AgNO_3$ is added to a 1.00 L saturated solution of Ag_2CrO_4. Assuming no volume change, does $[CrO_4^{2-}]$ increase, decrease, or remain the same? Justify your answer.

   In a saturated solution of Ag_3PO_4 at 25°C, the concentration of $Ag^+(aq)$ is 5.3×10^{-5} M. The equilibrium-constant expression for the dissolving of $Ag_3PO_4(s)$ in water is shown below.

   $$K_{sp} = [Ag^+]^3[PO_4^{3-}]$$

   (e) Write the balanced equation for the dissolving of Ag_3PO_4 in water.

   (f) Calculate the value of K_{sp} for Ag_3PO_4 at 25°C.

   (g) A 1.00 L sample of saturated Ag_3PO_4 solution is allowed to evaporate at 25°C to a final volume of 500. mL. What is $[Ag^+]$ in the solution? Justify your answer.

Ksp FRQ #3

1. Answer the following questions that relate to solubility of salts of lead and barium.

   (a) A saturated solution is prepared by adding excess $PbI_2(s)$ to distilled water to form 1.0 L of solution at 25°C. The concentration of $Pb^{2+}(aq)$ in the saturated solution is found to be 1.3×10^{-3} M. The chemical equation for the dissolution of $PbI_2(s)$ in water is shown below.

   $$PbI_2(s) \rightleftarrows Pb^{2+}(aq) + 2\ I^-(aq)$$

   (i) Write the equilibrium-constant expression for the equation.

   (ii) Calculate the molar concentration of $I^-(aq)$ in the solution.

   (iii) Calculate the value of the equilibrium constant, K_{sp}.

   (b) A saturated solution is prepared by adding $PbI_2(s)$ to distilled water to form 2.0 L of solution at 25°C. What are the molar concentrations of $Pb^{2+}(aq)$ and $I^-(aq)$ in the solution? Justify your answer.

   (c) Solid NaI is added to a saturated solution of PbI_2 at 25°C. Assuming that the volume of the solution does not change, does the molar concentration of $Pb^{2+}(aq)$ in the solution increase, decrease, or remain the same? Justify your answer.

   (d) The value of K_{sp} for the salt $BaCrO_4$ is 1.2×10^{-10}. When a 500. mL sample of 8.2×10^{-6} M $Ba(NO_3)_2$ is added to 500. mL of 8.2×10^{-6} M Na_2CrO_4, no precipitate is observed.

   (i) Assuming that volumes are additive, calculate the molar concentrations of $Ba^{2+}(aq)$ and $CrO_4^{2-}(aq)$ in the 1.00 L of solution.

   (ii) Use the molar concentrations of $Ba^{2+}(aq)$ ions and $CrO_4^{2-}(aq)$ ions as determined above to show why a precipitate does not form. You must include a calculation as part of your answer.